Classical Cipher

Classical Substitution Ciphers

The two basic building blocks of all encryption techniques are substitution and transposition.

A **substitution technique** is one in which the letters of plaintext are replaced by other letters or by numbers or symbols.

If the plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with ciphertext bit patterns

Mono-alphabetic Cipher

A **mono-alphabetic substitution cipher** is a type of substitution cipher where each letter in the plaintext is replaced by a corresponding letter in the ciphertext.

The key feature of this cipher is that the substitution **remains consistent throughout the entire message—each** letter from the plaintext alphabet always maps to the same letter in the ciphertext alphabet.

Caesar Cipher

The Caesar cipher, also known as a **shift cipher**, is one of the simplest and most widely known encryption techniques.

It is a type of substitution cipher where each letter in the plaintext is shifted by a certain number of places **down or up** the alphabet.

The first attested use in military affairs of one was by Julius Caesar, described by him in Gallic Wars.

How Caesar Cipher Works

The Caesar cipher operates on the **alphabetic text**, replacing each letter in the plaintext with a letter found by moving a certain number of positions to the **right or left** within the alphabet.

Encryption Process

Plaintext: The original message you want to encrypt.

Key: The number of positions each letter in the plaintext will be shifted.

Ciphertext: The encrypted message.

How Caesar Cipher Works

Let’s break it down step by step:

- You pick a key (which is the number of shifts).
- For each letter in the plaintext, you shift it by the value of the key.

For example, with a key of 3:

- **A** becomes **D**
- **B** becomes **E**
- **C** becomes **F** ...and so on.

How Caesar Cipher Works

Example:

Let's encrypt the word **HELLO** with a key of 3.

Shift each letter by 3 positions:

1. H → K
2. E → H
3. L → O
4. L → O
5. O → R

Ciphertext: KHOOR

How Caesar Cipher Works

Decryption Process

To decrypt the message, the same process is reversed. You subtract the key from each letter to return to the original plaintext.

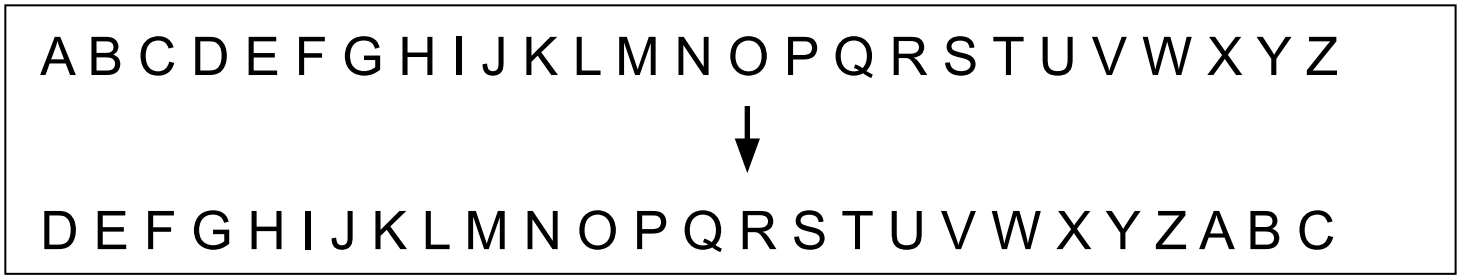
For example, to decrypt **KHOOR** with a key of 3:

- K → H
- H → E
- O → L
- O → L
- R → O

Thus, the decrypted plaintext is **HELLO**.

How Caesar Cipher Works

Caesar Cipher is a method in which each letter in the alphabet is rotated by three (key) letters as



Let us try to encrypt the message

"Attack at Dawn" using key 4

Class Work

Each student will exchange a secret message with his/her closest neighbor about some other person in the class and the neighbor will decipher it.

Mathematical Representation

The Caesar cipher can be represented mathematically. Let's assume:

P = Plaintext character (as a number from 0 to 25, where A = 0, B = 1, ..., Z = 25)

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12

n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

C = Ciphertext character

K = Key (number of shifts)

The encryption function is: **C = (P+K) mod 26**

The decryption function is: **P = (C−K) mod 26**

Example:

If **P = 7** (representing **H**) and **K = 3**, the ciphertext would be: C = (7+3) mod 26 = 10 corresponds to **K** in the alphabet.

Class Work

howdy (7,14,22,3,24) encrypted using key *f* (ie a shift of 5) is?????

Requirements

Two requirements for secure use of symmetric encryption:

- A strong encryption algorithm
- A secret key known only to sender / receiver

- Mathematically have:

  $Y = E(K, X)$

  $X = D(K, Y)$

- Assume encryption algorithm is known
- **Implies a secure channel to distribute key**

Strengths of Caesar Cipher

Despite being a simple cipher, the Caesar cipher has a few strengths:

1. **Easy to Understand and Implement**: It's an excellent teaching tool to explain basic cryptographic principles.
2. **Fast Encryption/Decryption**: Due to its simplicity, it requires minimal computational power.
3. **Low Storage and Memory Requirements**: No special hardware or complex algorithms are needed.
4. **Basis for Other Ciphers**: The Caesar cipher is the foundation for more complex encryption methods and ciphers.

Weaknesses of Caesar Cipher

The Caesar cipher is not secure in modern cryptography. Its weaknesses include:

1. **Limited Key Space**: There are only 25 possible keys (since shifting by 26 brings you back to the same alphabet). An attacker can try all possible shifts (brute-force attack) and easily break the cipher.
2. **Letter Frequency Analysis**: Since the Caesar cipher preserves letter frequency, it's vulnerable to frequency analysis. In English, certain letters (like 'E' and 'T') appear more frequently, and these patterns can be exploited to crack the cipher.
3. **Predictable Shifting**: Once the key is discovered, the entire cipher becomes exposed, making it easy to decrypt.
4. **No Use of Modern Cryptographic Techniques**: It does not provide confidentiality, integrity, or authentication as required in modern communication standards.

Breaking the Caesar Cipher

Let's break a Caesar cipher-encrypted message using **brute force** or **frequency analysis**:

- **Example of Brute Force Attack:**
- Given the ciphertext: **LIPPS**, we try all possible shifts:
- With key 1: **KHOOR**
- With key 2: **JGNNQ**
- With key 3: **IFMMP**
- With key 4: **HELLO** (we recognize this as valid English text).
- Therefore, the key is 4, and the plaintext is **HELLO**.

Example of Frequency Analysis:

- If we know that the letter **E** is the most common in English, we can look for the most frequent letter in the ciphertext and assume it maps to **E**. We then calculate the shift and decrypt the rest of the message.

Improving the Caesar Cipher

To improve the Caesar cipher and address its weaknesses, more complex encryption techniques can be applied, such as:

1. **Vigenère Cipher**: Uses a keyword to vary the shift amount at each position in the plaintext.
2. **Modern Cryptographic Algorithms**: Algorithms like AES (Advanced Encryption Standard) use much more complex and secure methods for encryption.

Monoalphabetic Cipher

The Caesar cipher isn't very secure because it only has 25 possible keys. To make the encryption much stronger, we can use a method called **arbitrary substitution.**

Before explaining, let's define a **permutation**. A permutation is a specific arrangement of a set of elements where each element appears exactly once. For example, if you have a set S={a,b,c} the possible arrangements (permutations) of these elements are:

abc, acb, bac, bca, cab, cba

Monoalphabetic Cipher

A permutation of the alphabet is used to create the substitution. A permutation is simply a rearrangement of the letters in the alphabet. For possible ways to arrange these letters, which gives a huge number of possible keys. This makes the cipher much stronger than something like a Caesar cipher example, if you have 26 letters, there are 26! (factorial).

Example:

Plaintext Alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ

Ciphertext Alphabet: YHKQGXPJWFNEDBMZTOCIVLASRU

Here, the permutation is such that:

$$A \rightarrow Y,\ B \rightarrow H,\ C \rightarrow K,\ D \rightarrow Q\ \ldots\ Z \rightarrow U$$

Monoalphabetic Cipher

Example in Use:

Plaintext: “CRYPTOGRAPHY"
Ciphertext: "KIUDCZIXYDFU“

In this example, "C" in the plaintext always maps to "K" in the ciphertext, "R" always maps to "I," and so on, following the permutation.

Monoalphabetic Cipher

```
Plain:  abcdefghijklmnopqrstuvwxyz
Cipher: DKVQFIBJWPESCXHTMYAUOLRGZN

Plaintext:  ifwewishtoreplaceletters
Ciphertext: WIRFRWAJUHYFTSDVFSFUUFYA
```

Decrypt this one?

Cipher text: BHHQ SOVE

Plaintext: ?

Monoalphabetic Cipher Security

Now have a total of 26! = 4 x 10^{26} keys.

With so many keys, might think is secure but would be **!!!WRONG!!!**

Problem is language characteristics

Example:

Imagine the word "HELLO" appears often in a message. Even with different keys, "H" will always be replaced by the same letter, "E" by another, and so on. A smart attacker can still figure out what the message says by looking for these common patterns in the language.

Language Redundancy and Cryptanalysis

- Human languages are **redundant**
- Letters are not equally commonly used
- In English E is by far the most common letter
- followed by T,R,N,I,O,A,S
- Other letters like Z,J,K,Q,X are fairly rare

English Letter Frequencies

